IEEE ENIT Robotics & Automation Society

IEEE Sensors Council

IEEE ENIT STUDENT BRANCH
JOINT CHAPTER

TRSYP

IEEE RAS SEN ENIT SBJC TEAM

TRSYP 2.0 Technical Challenge

Smart Autonomous Vehicle - EmbodiedAI

TEAM OPTIMUS

TABLE OF CONTENTS

1- OPTIMUS TEAM

2 - PROJECT OVERVIEW

3 - SYSTEM ARCHITECTURE

4 - HARDWARE COMPONENTS

5 - SYSTEM MODELS

6 - IMPLEMENTATION

MEET OUR BRIGHT MINDS



WAJDI DRIDI



REBEH MEZRIGUI



WASSIM TOUKEBRI



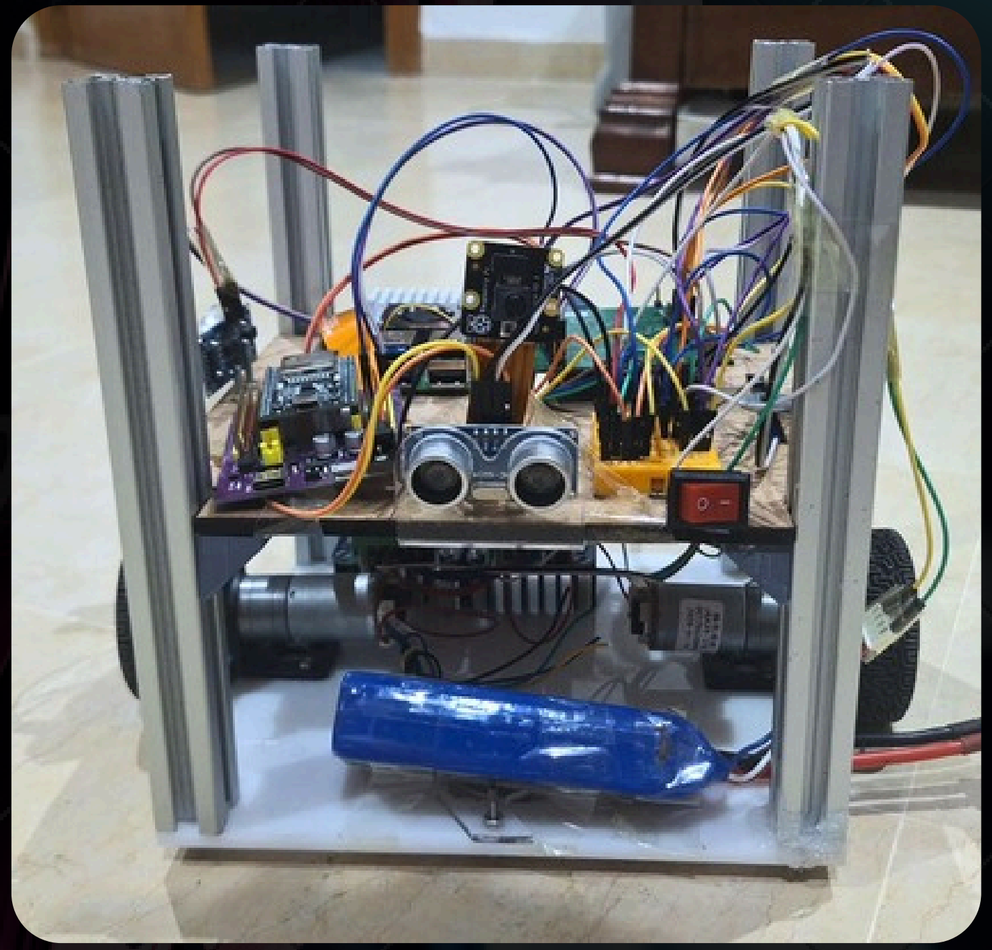
SOUHA KHALFAOUI

PROJECT OVERVIEW

Our project aims to create a **Hybrid Robotic System** capable of operating in **both autonomous and manual modes** , integrating advanced **real-time control technologies and artificial intelligence.**

The robot integrates four main modules :

- **Real-Time Control Module**
- **Autonomous Intelligence Module**
- **Communication and Telemetry Module**
- **Power Management Module**



SYSTEM ARCHITECTURE

MQTT

Networked, real-time communication between subsystems for data exchange and telemetry.

UART

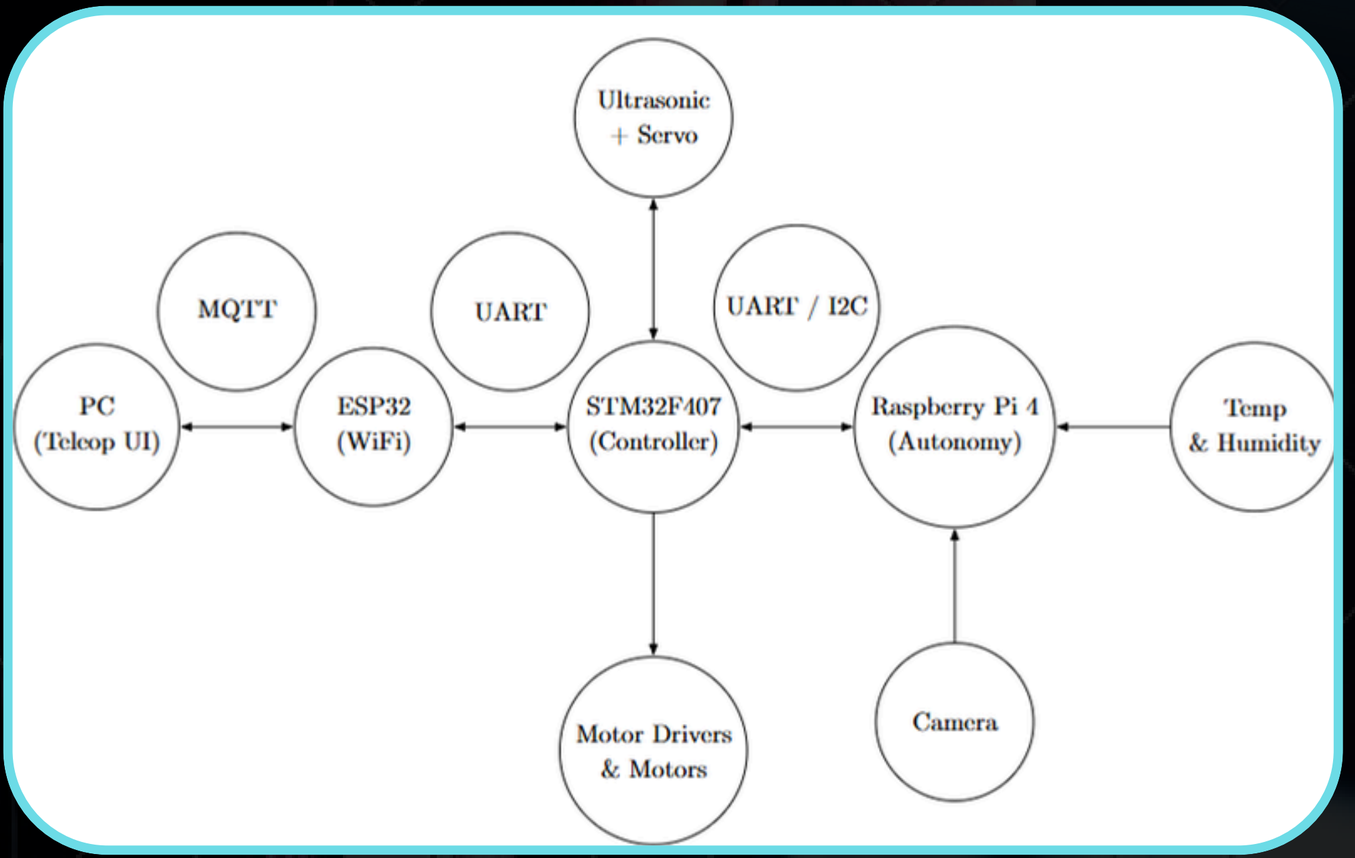
Reliable hardware serial link for commands and data.

I2C

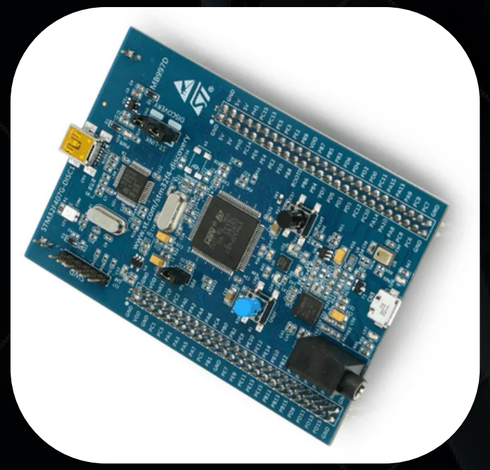
Short-range communication between MCU and sensors.

PWM

Motor speed/direction control via duty cycle.



HARDWARE COMPONENTS



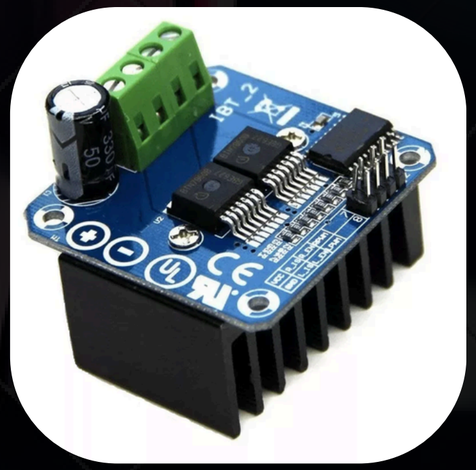
STM32F407G-DISC1 Board



3 HC-SR04 Ultrasonic Sensors



2 JGA25-370 DC Motors with Encoders



2 BTS7960 Motor Driver Modules

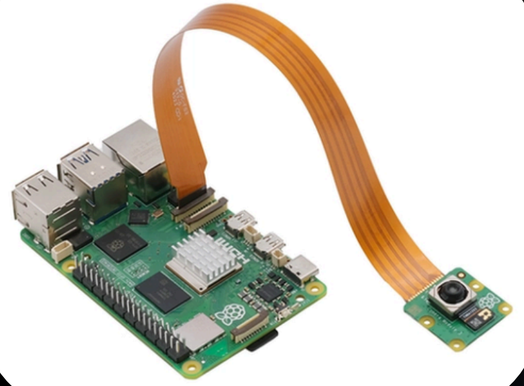
HARDWARE COMPONENTS



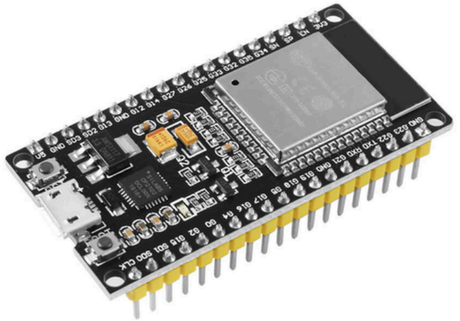
LiPo Battery 11.1V



Power Bank

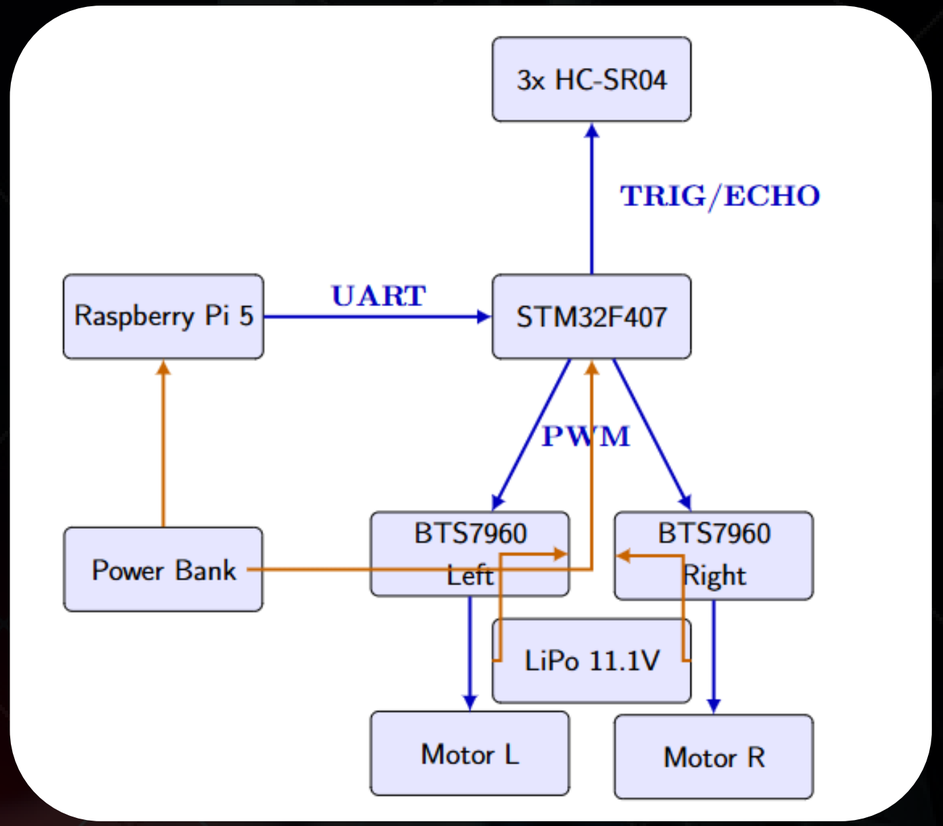


Raspberry Pi + Camera + Nappe



ESP32 Dev Kit

SYSTEM MODELS

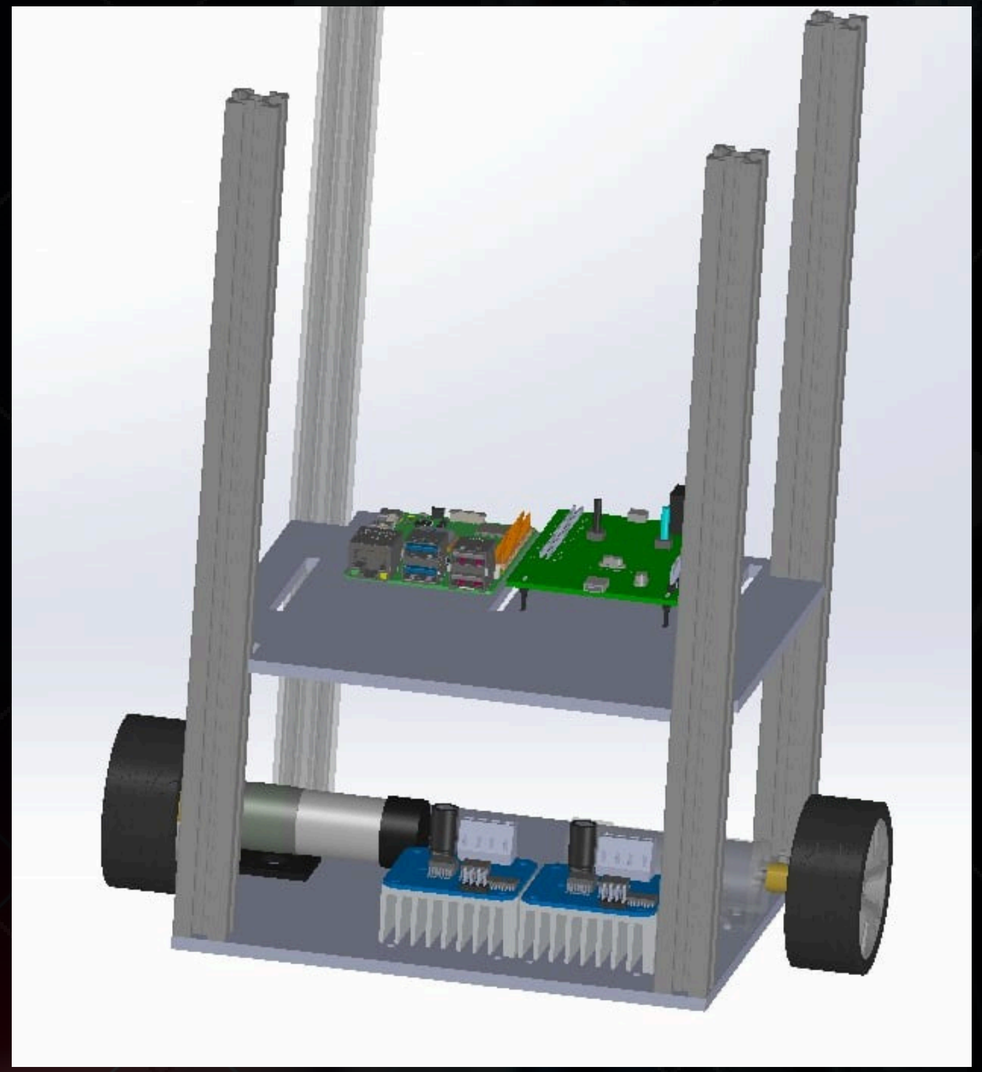


Control Model

Differential drive control:
Forward → vL = vR
Turn Left → vL = −v, vR = +v
Turn Right → vL = +v, vR = −v

Obstacle thresholds:
Front: 50 cm
Sides: 20 cm

Speed = speedmin + d_1 × (speedmax – speedmin) / 100
Stuck detection : Backup + turn if all sensors detect obstacle



3D CAD Model (SolidWorks)

The robot was **modeled and assembled in SolidWorks before physical construction**.

The **3D model** provided a clear view of the **chassis, sensor layout, motor positioning, and wiring paths.**

This **virtual prototype** allowed us to :

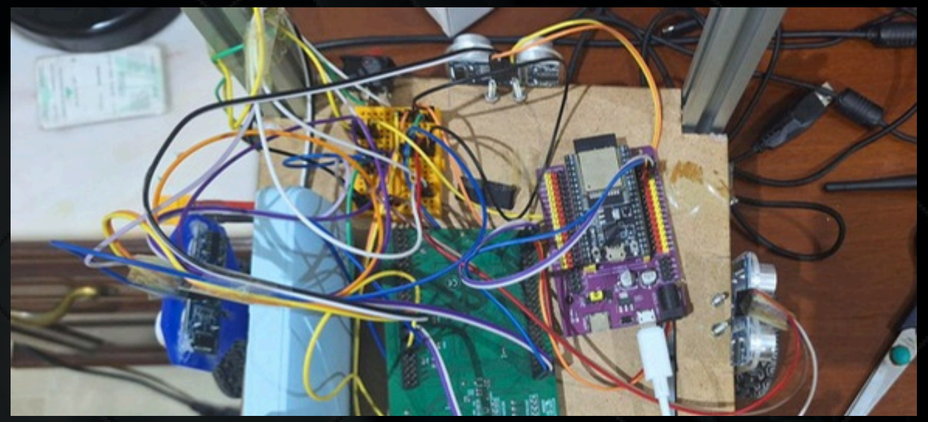
- Optimize weight distribution and component placement.
- Detect and solve integration issues early.
- Ensure a clean and efficient mechanical design.

IMPLEMENTATION

1 Hardware Integration

Steps:

1. **Mount motors & wheels**
2. **Install ball caster**
3. **Place ultrasonic sensors**
4. **Connect BTS7960 drivers**
5. **Wire power system**
6. **Test & calibrate motion**



2 Software Architecture

STM32F407 : Real-Time Control

FreeRTOS-based architecture

Main tasks : Motor control, encoder, sensors, communication, safety, telemetry

Key features : PID regulation, real-time data, fault protection

IMPLEMENTATION

Raspberry Pi 5 – AI-Powered Risk Prediction

- AI navigation & path planning
- MQTT communication
- Node-RED dashboard
- Data logging & analysis

3 Navigation Algorithm:

- Sensors: 3 ultrasonic (front, left, right)
- Thresholds:
  - Front → 50 cm
  - Sides → 20 cm
- Behavior:
  - Stop → Backup → Turn → Go
  - Adaptive speed (180–250 PWM)
  - Self-correction if stuck

4 Communication System

- **MQTT: Mosquitto broker for data exchange**
- **Node-RED: Dashboard for monitoring & control**

CONCLUSION

Our project resulted in a functional prototype that combines embedded control, basic AI navigation, and a stable mechanical design.

It demonstrates the application of theoretical knowledge in a practical system and highlights areas for future improvement in autonomy and performance.

THANK YOU FOR YOUR ATTENTION

